

Skeletal System of Vertebrates

→ Endoskeleton

→ Two types



Bones

Cartilage

Cartilage

→ Specialized kind of connective tissue.

→ provides site for muscle attachment

→ Support

↳ Cartilage Cartilage

- Specialized kind of connective tissue.
- provides site for muscle attachment
- Support
- helps in movement.
- Transmits the force of muscular contraction from one part of body to other.
- Composed of

→ helps in movement.

→ Transmits the force of muscular contraction from one part of body to other.

Composed of

↓
Chondrocytes

↓
fibers

↓
cellular matrix

Bones

→ kind of connective tissue.

✓ Rigid

Chondrocytes

Fibers

Cellular matrix

Bones

→ Kind of connective tissue.

→ Rigid

→ provides site for muscle attachment.

→ helps in movement.

→ Transmit force from one part
to other.

→ protect internal organs

→ Store Ca^{+} + P

→ formation of blood cells.

Its rigid because

Its homogenous organic substance has inorganic substances like P, Ca^{+} .

Bones → present → minute chambers → called lacunae.

↓
arranged in concentric rings
around central canal.

(Haversian System)

Cells communicate with nearby
cells by means of cellular
process passing through small
channels → Canaliculi.

Skeleton of Fishes

→ Both exoskeleton and endoskeleton appears in fishes.

→ fishes need ~~any~~ support in water.

→ Featureless vertebral:-

→ Hagfish
→ Lamprey } No vertebral column.

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Hagfish:-

Only Notochord

Lamprey:-

Cartilaginous blocks around the notochord.

Jawed fishes:-

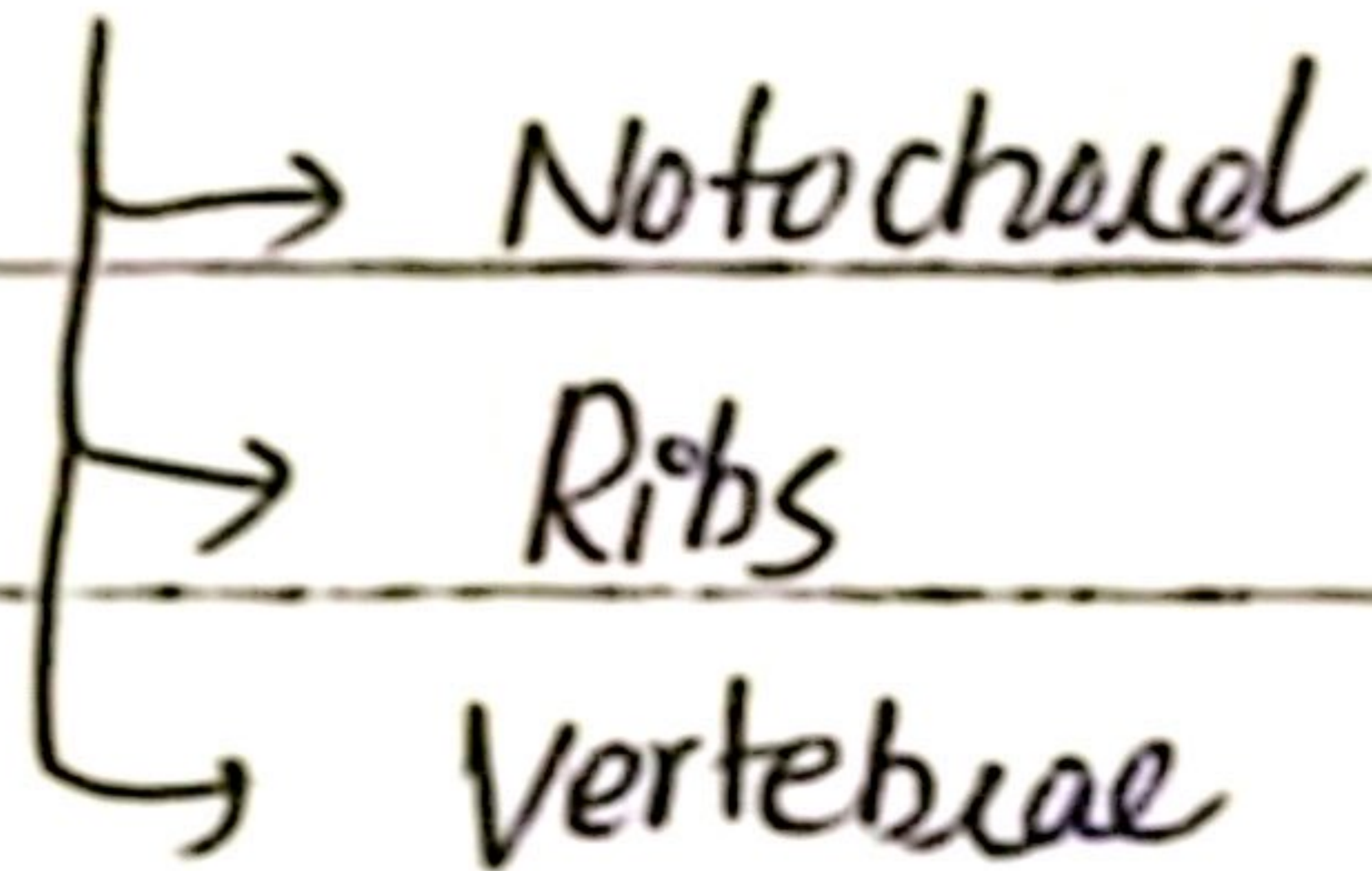
→ Vertebrates

Notochord.

Jawed fishes:-

→ Vertebrae

→ axial skeleton



Muscles attach to axial skeleton.

The Skeleton of Tetrapods

→ 200 my ago.

→ supportive skeleton → on land.

Characteristics:

→ Intervertebral disk that

articulates with adjoining vertebrae.

→ Intervertebral disk

↳ hold the vertebrate column

together.

- hold the vertebrate column
- Absorb shock
- Provide joint mobility.

Bone → replaced cartilage in ribs.

rigidity

Human Endoskeleton:-

^{rigidly}
∴ Human Endoskeleton :-

Axial skeleton

Appendicular skeleton

→ ribs

→ sternum

→ vertebrae

Pectoral girdle

Pelvic girdle